

Fake News Classification Report

1. Introduction

The spread of fake news has become a significant challenge in today’s digital world. With the constant flow of news articles published daily, it is becoming harder to distinguish between credible and misleading information. This creates a need for systems that can automatically classify articles as true or fake, helping reduce misinformation and protect public trust. This project focuses on developing a machine learning system to automatically classify news articles as **real** or **fake**, leveraging semantic features for better generalization.

2. Problem Statement

Traditional keyword- or frequency-based methods struggle to capture the deeper **contextual meaning** of words. Fake news often uses language in subtle ways, making shallow models unreliable.

We address this by using **Word2Vec embeddings**, which transform words into dense 300-dimensional vectors that encode meaning based on context. These embeddings allow classifiers to detect semantic similarity and patterns in news articles.

3. Dataset

The dataset consists of labeled news articles with the target variable:

- 1 = True news
- 0 = Fake news

df_true.head(2)			
	title	text	date
0	As U.S. budget fight looms, Republicans flip their fiscal script	1.5trillion, tobepaidoffbyfuturegenerations, to financetazcuts forcorporationsandtherich. \nThisisoneoftheleast... fiscallyresponsiblebillswe'veeverseemp	
1	U.S. military to accept transgender recruits on Monday; Pentagon	WASHINGTON (Reuters) - Transgender people will be allowed for the first time to enlist in the U.S. military starting on Monday as ordered by federal courts, the Pentagon	
df_fake.head(2)			
0	Donald Trump Sends Out Embarrassing New Year's Eve Message; This is Disturbing	Donald Trump just couldn't wish all Americans a Happy New Year and leave it at that. Instead, he had to give a shout out to his enemies, haters and the very dishonest fake news media. The former reality show star had just one job to do and he couldn't do it. As our Country rapidly grows stronger and smarter, I want to wish all of my friends, supporters, enemies, haters, and even the very dishonest Fake News Media, a Happy and Healthy New Year, President Angry Pants tweeted. 2018 will be a great year for America! As our Country rapidly grows stronger and smarter, I want to wish all of my friends, supporters, enemies, haters, and even the very dishonest Fake News Media, a Happy and Healthy New Year. 2018 will be a great year for America! Donald J. Trump (@realDonaldTrump) December 31, 2017Trump's tweet went down about as well as you'd expect. What kind of president sends a New Year's greeting like this despicable, petty, infantile gibberish? Only Trump! His lack of decency won't even allow him to rise above the gutter long enough to wish the American citizens a happy new year! Bishop Talbert Swan (@TalbertSwan) December 31, 2017no one likes you Calvin (@calvinstowell) December 31, 2017Your impeachment would make 2018 a great year for America, but I'll also accept regaining control of Congress. Miranda Yaver (@mirandayaver) December 31, 2017Do you hear yourself talk? When you have to include that many people that hate you you have to wonder? Why do they all hate me? Alan Sandoval (@AlanSandoval13) December 31, 2017Who uses the word Haters in a New Year's wish?? Marlene (@marlene399) December 31, 2017You can't just say happy new year? Koren politt (@Korencarpenter) December 31, 2017Here's Trump's New Year's Eve tweet from 2016.Happy New Year to all, including to my many enemies and those who have fought me and lost so badly they just don't know what to do. Love! Donald J. Trump (@realDonaldTrump) December 31, 2016This is nothing new for Trump. He's been doing this for years. Trump has directed	December 31, 2017

WordCloud of True v/s Fake News from the dataset-



Data preprocessing included:

- Cleaning text (punctuation, stopwords, lowercase normalization)
- Tokenization and embedding using **pre-trained Word2Vec (GoogleNews-vectors-300)**

4. Methodology

We implemented the following pipeline:

1. **Data Preprocessing** – tokenization, cleaning, embedding.

Lowercasing

- Convert all text to lowercase to ensure consistency.

Remove text in square brackets

- Eliminate irrelevant references, citations, or annotations.
- Example: "This news [Reuters]" → "This news"

Remove punctuation

- Strip commas, periods, symbols, etc., to simplify tokenization.
- Example: "fake-news!!!" $\rightarrow$ "fakenews"

Remove words containing numbers

- Exclude noisy tokens like "covid19", "2020update".

POS Tagging & Lemmatization

- Apply **Part-of-Speech (POS) tagging** to each token.
- Keep only **nouns** (NN, NNS) since they often represent key entities in fake news detection (e.g., “*president*”, “*election*”).
- Lemmatize to reduce words to their base form:

"reports" → "report", "stories" → "story"

2. **Feature Extraction** – Word2Vec embeddings converted text into semantic vectors.
After preprocessing, the cleaned news text is transformed into Word2Vec embeddings. Word2Vec converts each word into a dense vector (300 dimensions) that captures semantic meaning.
Words appearing in similar contexts (e.g., “president”, “leader”) end up with vectors that are close to each other in space.
Each news article is represented by averaging the vectors of its words.
This produces a single semantic vector representation for the whole article.
Unlike keyword counts (TF-IDF), Word2Vec understands context and meaning, allowing classifiers to generalize better even when different words are used to express the same idea.
3. **Model Training** – On this we trained Logistic Regression, Random Forest, and Decision Tree classifiers.
4. **Evaluation** – measured performance with Accuracy, Precision, Recall, and F1-score.

Model Evaluation

Logistic Regression Model

7.1.2 Calculate and print accuracy, precision, recall and f1-score on validation data [5 marks]

```
] : ## Calculate and print accuracy, precision, recall, f1-score on predicted labels
accuracy = accuracy_score(y_val_array, y_val_pred)
precision = precision_score(y_val_array, y_val_pred, average='weighted')
recall = recall_score(y_val_array, y_val_pred, average='weighted')
f1 = f1_score(y_val_array, y_val_pred, average='weighted')
print(f"Accuracy on validation set: {accuracy:.4f}")
print(f" Precision: {precision:.4f}")
print(f"Recall:      {recall:.4f}")
print(f"F1-score:     {f1:.4f}")
```

```
Accuracy on validation set: 0.9538
Precision: 0.9539
Recall:    0.9538
F1-score:  0.9538
```

Decision Tree Model

```

## Calculate and print accuracy, precision, recall, f1-score on predicted labels
accuracy = accuracy_score(y_val_array, y_val_pred_dt)
precision_dt = precision_score(y_val_array, y_val_pred_dt, average='weighted')
recall_dt = recall_score(y_val_array, y_val_pred_dt, average='weighted')
f1_dt = f1_score(y_val_array, y_val_pred_dt, average='weighted')

print(f" Decision Tree Accuracy on validation set: {accuracy:.4f}")
print(f"Precision: {precision_dt:.4f}")
print(f" Recall:      {recall_dt:.4f}")
print(f"F1-score:     {f1_dt:.4f}")

```

```

Decision Tree Accuracy on validation set: 0.8732
Precision: 0.8736
Recall:    0.8732
F1-score:  0.8730

```

Random Forest Model –

```

: ## Calculate and print accuracy, precision, recall, f1-score on predicted labels
accuracy_rf = accuracy_score(y_val_array, y_val_pred_rf)
precision_rf = precision_score(y_val_array, y_val_pred_rf, average='weighted')
recall_rf = recall_score(y_val_array, y_val_pred_rf, average='weighted')
f1_rf = f1_score(y_val_array, y_val_pred_rf, average='weighted')

```

```

: # Classification Report
# Step 6: Print Classification Report
print("\n 📊 Classification Report (Random Forest):")
print(classification_report(y_val_array, y_val_pred_rf))

```

```

📊 Classification Report (Random Forest):
              precision    recall  f1-score   support

     0       0.94       0.95       0.95        6978
     1       0.95       0.94       0.94        6498

 accuracy          0.94          0.94          0.94       13476
 macro avg         0.94          0.94          0.94       13476
 weighted avg      0.94          0.94          0.94       13476

```

5. Results

The performance of models is summarized below:

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.95	0.95	0.95	0.95

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.94	0.94	0.94	0.94
Decision Tree	0.87	0.87	0.87	0.87

Key finding: Logistic Regression consistently outperformed more complex models, proving that linear models can achieve excellent results with semantically rich embeddings.

6. Assumptions

- The dataset is representative of broader real-world news content.
- Word2Vec embeddings trained on Google News sufficiently generalize to the dataset.
- News articles are labeled accurately without bias.

7. Conclusion

The semantic classification approach with **Word2Vec + Logistic Regression** proved highly effective, achieving **95% accuracy**.

By leveraging embeddings, the model overcomes the limitations of keyword-based methods and captures the **meaning and intent** behind text. This enables a more robust and generalizable fake news detection system.

Future improvements may include:

- Using transformer-based embeddings (e.g., BERT, RoBERTa).
- Exploring ensemble models for better robustness.
- Evaluating model interpretability for real-world adoption.